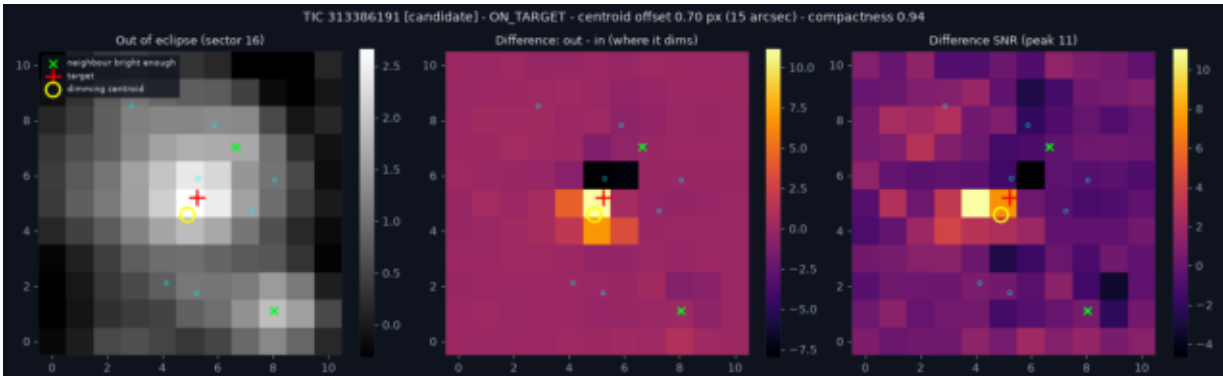
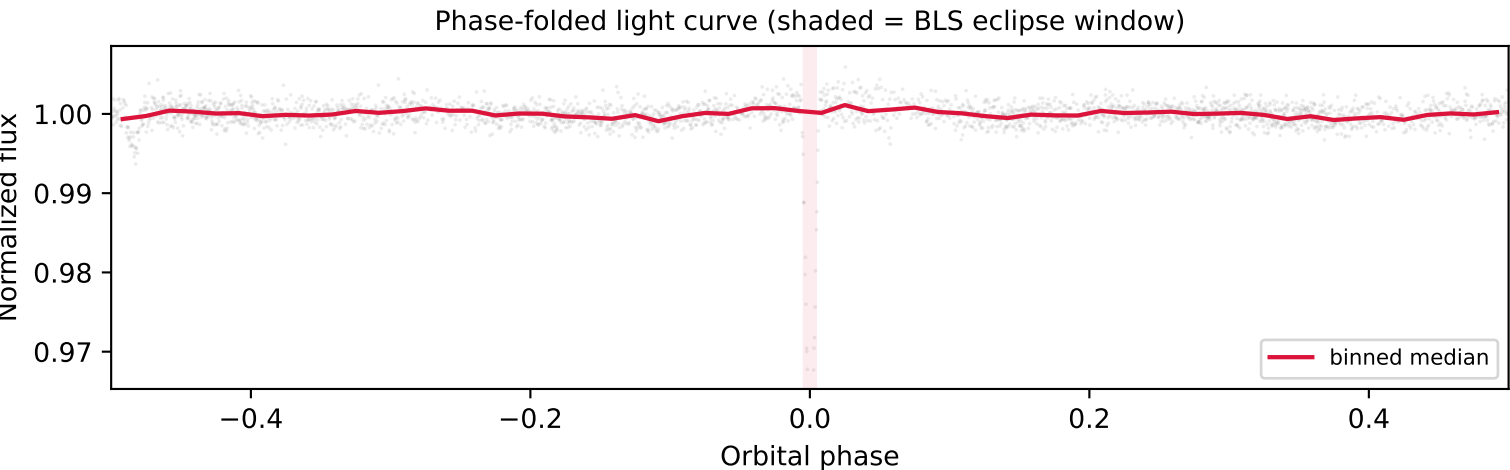


TIC 313386191

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS		CATALOG CROSS-MATCH (novelty)	
Period	20.362255 d (ALIASED, see below)	VSX	no match
-> corrected period:	40.724510 d	ASAS-SN	no match
Epoch (t0)	1750.0737 BTJD	Gaia eclipsing-binary table	no match
Duration	0.1700 d	Gaia variable classification	no match
Depth	3.316 % (33,164 ppm)	TESS EB catalog (Prsa+2022)	no match
SNR	121.0	ExoFOP TOI list	no match
INDEPENDENT CHECKS (f3i gates)		Verdict: NOVEL	
Eclipses observed	3 (gate >= 3)		
Robust depth ratio	1.10 (gate >= 0.50)		
Robust depth sigma	121.3 (gate >= 7.0)		
Odd/even z-score	8.62 (ALIAS FLAGGED)		
Secondary eclipse	detected, phase -0.48, 5.5 sigma		
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)			
Sector 16	light-curve dip in this sector: 9.7 sigma	t0 refined by	-0.4850 phase
Difference-image peak SNR	11.07 (gate >= 5.0)	Compactness	0.94 (gate >= 0.50)
Dimming centroid offset	0.70 px (15 arcsec)	(gate <= 1.0 px, i.e. on the target)	
Nearest bright-enough neighbour in cutout:	TIC 313386192		
	3.02 px away, dTmag +2.70 - farther from the dimming than the target		



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 8.62 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 40.724510 d, in any write-up.